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RESEARCH ARTICLE

A method of requirements elicitation and analysis for Global Software Development

Naveed Ali | Richard Lai

Department of Computer Science and Information Technology, La Trobe University, Melbourne, Vic 3083, Australia

Correspondence
Richard Lai, Department of Computer Science and Information Technology, La Trobe University, Melbourne, Vic 3083, Australia.
Email: r.lai@latrobe.edu.au

Abstract

To perform requirements elicitation and analysis, effective communication and collaboration between stakeholders are necessary. Global Software Development (GSD), where software teams are located in different parts of the world, has become increasingly popular. However, geographical distance, cultural diversity, differences in time zones, and language barriers create difficulties for GSD stakeholders in engaging in effective communication. Taking into consideration the factors involved in GSD, previous research showed that the ways by which requirements are gathered and analyzed for collocated software development cannot be used effectively for GSD. Thus, in this paper, we present a method of requirements elicitation and analysis for GSD. The method consists of 4 stages: (1) data collection; (2) educating stakeholders about GSD issues; (3) post-education assessment; and (4) requirements elicitation and analysis. Past researchers used student groups in a university environment to play the roles of stakeholders in experiments in GSD studies. Likewise, we preliminarily validate our method by applying it to a case study of an online shopping system, where the roles of client, requirements engineer, project analyst, and designers were played by a group of students.

KEYWORDS

distributed teams, Global Software Development, requirements analysis, requirements elicitation

1 | INTRODUCTION

Requirements engineering (RE) is a systematic process of gathering, analyzing, validating, and organizing requirements obtained from the stakeholders of a computer system project.[1–3] Its success hinges on the practice and performance of these RE activities; hence, it is an important part of project development. Requirements engineering also includes resolving conflicts regarding completeness and inconsistencies.[4] There are several techniques available for requirements elicitation and analysis. Each of these techniques has advantages and disadvantages and can be used in specific domains. The techniques commonly used for requirements elicitation include focus groups,[5] interviews,[6,7] scenarios and use cases,[8,9] joint application development,[10] brainstorming,[11,12] and user stories.[13] Likewise, the widely used requirements analysis techniques are card sorting,[14] affinity diagrams,[15] storyboards,[16] and function allocation.[17]

Global Software Development (GSD) is multisite software development with software teams scattered across different places around the world.[18,19] To gain the benefit of lower costs, faster and improved software development and services, GSD is a good solution for many organizations.[20,21] The number of organizations engaged in GSD is increasing, and a considerably large amount of funds have been poured into it.[22] However, GSD has introduced issues for stakeholders that are not present in software projects developed in the same location (collocated projects).[23–25] Because of development teams being in different geographical locations, differences in cultures and time zones[26,27] adversely impact communication and coordination processes.[28–31] Consequently, the frequency of communication, coordination, and trust between the development teams decreases.[32] In addition, dissimilar approaches and processes of software development, technical capabilities of remotely distributed team members, and less visibility of development work performed at different sites create additional issues for stakeholders to tackle. Of the major issues to successful GSD, requirements elicitation and analysis are of significance.

To perform requirements elicitation and analysis, effective communication and collaboration between stakeholders are necessary. Preceding research has mentioned different problems, in which a lack of communication between stakeholders is a significant issue.[33] Geographical distance, cultural diversity, differences in time zones, and language barriers create difficulties for GSD stakeholders in engaging in effective communication.[34] Traditional approaches to requirements elicitation and analysis do not consider factors such as distance, time zones, cultural diversity, communication, and language barriers. As a result, the ways by which requirements are captured and analyzed in collocated software development cannot be used effectively in a globally distributed environment.[35,36] To date, there are very few research papers that describe work done on requirements elicitation and analysis for GSD. In this paper, we present GSD-RE, a requirements elicitation and analysis method for GSD, especially for those groups of stakeholders who take part in GSD for the first time or are not aware of the fundamental aspects and issues of RE in GSD. Our method facilitates stakeholders in accomplishing the following: (1) gaining awareness of the societal, temporal, geographical, and lingual aspects of GSD at an early stage, which helps them during the processes of requirements elicitation and analysis; (2) the consideration of nonfunctional requirements with respect to different time zones and increased distance helps stakeholders think about the underlying requirements in a broader domain, which are often neglected in collocated software development; and (3) devising a quality set of requirements.

Past researchers used student groups in a university environment to play the roles of stakeholders in experiments in GSD studies.[37–40] Likewise, we preliminarily validate our method by applying it to a case study of an online shopping system (OSS), where the roles of client, requirements engineer, project analyst, and designers were played by a group of students. Throughout the paper, we refer to these groups of people as "stakeholders."

2 | OUR REQUIREMENTS ELICITATION AND ANALYSIS METHOD

The requirements elicitation and analysis method is divided into 4 stages: (1) data collection; (2) educating stakeholders about GSD issues; (3) post-education assessment; and (4) requirements elicitation and analysis. The method is depicted in Figure 1, whereas the details of the roles and responsibilities performed by the GSD stakeholders can be found in Table 1.

The method starts by collecting basic information on stakeholders in stage 1. Based on the collected information, potential conflicts regarding cultural diversity, communication and coordination, and time zones will be identified in stage 1 and further addressed by educating stakeholders in stage 2. In stage 3, post-education assessment will be performed to evaluate the knowledge gained by stakeholders. Considering the outcomes of post-assessment evaluations, decisions will be made about whether further explanation of certain issues will be required; otherwise, the elicitation and analysis process will begin in stage 4 by asking clients to write their WANTS regarding system requirements in user story format.

The information obtained will be used to identify WH questions (ie, what does the client want, why and when it is required, and who will use it) about the functional and nonfunctional requirements of the prospective systems. The pool of WH questions will be analyzed, and a set of WH diagrams will be generated. From the information presented in the WH diagrams, scenarios will be identified that define the different aspects of requirements in a detailed manner. After the

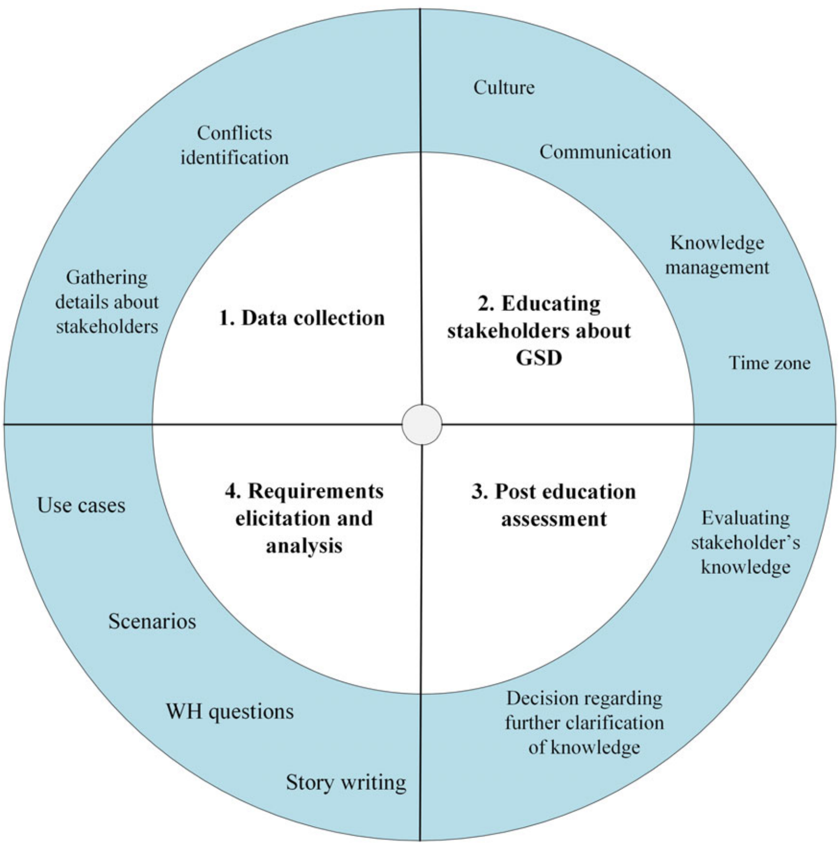

FIGURE 1 Method of requirements elicitation and analysis

TABLE 1 Roles played by the stakeholders

Role	Task	Responsibilities
Client	Information provider	Defines user software requirements
Analyst	Implements proposed RE method	Performs data collection (phase 1), stakeholder's education (phase 2), post-education assessment (phase 3), and requirements elicitation and analysis (phase 4)
Requirements engineer	Requirements elicitation and analysis	Performs requirements elicitation and analysis (phase 4)
Designers	Requirements analysis	Perform requirements analysis

identification of possible scenarios, use cases will be defined for them. It is a recursive process that will continue until no further exploration is needed.

Wiegers and Beatty[41] stated that user stories, scenarios, and use cases are used to facilitate different types of GSD stakeholders in understanding software requirements and fill the gap between user research and design. User stories and scenarios are useful for user experience practitioners (eg, client and requirements engineers), because it is easy to communicate design information by using these approaches. However, use cases are defined to facilitate managers and analysts in understanding system response and user behaviors, so that they can identify what needs to be done.

2.1 | Data collection

To gather information about stakeholders, questionnaires are distributed to the GSD stakeholders (refer to Appendix S1).[42] Information on the following aspects will be obtained: (1) personal information—name, gender, and contact details; (2) language—first and second language; (3) culture and location—country of origin and residence; (4) time zones and working hours—current working hours during weekdays and weekend; (5) job description—role performed during RE; (6) collaborative tools—list of tools that are commonly used in the organization, preference for using different collaborative tools, and the level of comfort experienced using different tools; (7) learning capability—willingness to learn new techniques; and (8) awareness of GSD—previous involvement in GSD and awareness of social, infrastructural, and technological aspects of GSD. Afterwards, the outcomes collected from the questionnaires will be grouped to identify possible issues among the stakeholders regarding culture, time zones, communication, and knowledge management.

2.2 | Educating stakeholders about GSD

The success of a GSD project is affected by cultural differences,[26,27,42] time zones,[26,27] knowledge management,[43] and communication barriers.[32,42] As part of our RE method, the challenges introduced by these issues will be first explained to stakeholders. Later, strategies used to minimize the impacts of these issues will be explained.

2.2.1 | Culture

Culture plays an important role in the development of personality. One's culture has a determining influence on many aspects of an individual, such as education, religion, past experiences, personality, individual ways of thinking and working, social involvement, and affection. The impact of cultural differences is clearly evident in GSD, where stakeholders are in different locations around the world. Stakeholders from different cultures react differently to the same situation, resulting in many differences in their working styles of and problem solving approaches. Cultural misunderstandings can create difficulties globally distributed stakeholders who are working toward a common goal.[27] Some of the problems of culture in a GSD environment are due to (a) lack of trust and respect[44]; (b) differences in organizational setups, backgrounds, and working styles[45]; (c) lack of cross-cultural training[46]; and (d) communication breakdown between stakeholders.[47] To minimize the number of potential problems between stakeholders due to cultural differences, we adopt the cross-cultural model of Lewis.[48]

(a) Understanding the cultures of others: Stakeholders will be informed of the different aspects of other cultures and how these can affect the RE activities of a GSD project.

(b) Identifying the similarities between one's own culture and the cultures of others: Once the stakeholders establish an understanding of the essential aspects of the cultures of others, information will be provided to find the similarities and dissimilarities between different cultures. Attention will be paid to find commonalities in working styles and communication mechanisms between different cultures.

2.2.2 | Knowledge management

When stakeholders in GSD are located in different parts of the world, it is difficult for them to share knowledge because of differences in organizational structures, cultural setups, and language barriers.[43] In addition to differences in organizational structures and cultural setups, stakeholders in different geographical locations often use different sets of keywords to represent a similar concept.[49,50] In the absence of a proper communication channel, often project details are misinterpreted that can result in misunderstandings between stakeholders in later stages of the project.[51] To minimize the number of conflicts in knowledge management due to language barriers, we suggested the use of ontology.

2.2.3 | Knowledge clarification via ontology

To clarify the structure of knowledge and to minimize the number of ambiguities in terminologies, a set of keywords will be defined. Distinctions between these keywords will be made by classifying them at domain, generic, representational, task, and method levels of ontology.[52] Domain ontology is used to define the terminologies relevant to a particular type of domain (such as banking and education); generic ontology defines the generic set of terminologies within a specific domain (such as finance and trading); representational ontology defines the general representation of entities regardless of the information about what to represent (such as frame); task ontology describes the specific terms for a certain task; and method ontology describes the set of terminologies specific to a particular method of solving a problem. The glossary will be updated regularly with new keywords.

2.2.4 | Communication

The lack of face-to-face interaction and group awareness, and dissimilar communication standards often hamper the communication and coordination process in a GSD environment. Consequently, the frequency of communication between stakeholders is decreased that affects the software development process. Some of the problems of communication in a GSD environment are due to (a) time spent on identifying the right person with whom to communicate at the right time[53–55]; (b) the use of different terminologies to represent similar concepts[49,50]; (c) difficulties in knowledge sharing[56,57]; (d) and differences in time zones.[26,27] To minimize the problems that could occur because of communication barriers, we suggest the use of the following:

(a) Organizational ontology: To clarify organizational roles in a GSD environment, we use an organizational ontology. Ontologies have previously been used in organizations and consist of several mechanisms that mutually achieve coordination,[58] for instance, goals, positions, communication, authority, and work process. Different parts of an organization are categorized by the particular roles they play in accomplishing coordination with the earlier stated means. Therefore, an organizational ontology is defined as follows:

$$\text{Organization} \rightarrow \text{Departments}(\text{department}_1, \text{department}_2 \ldots \text{department}_n)$$

$$\text{Departments} \rightarrow \text{Managers}(\text{manager}_1, \text{manager}_2 \ldots \text{manager}_n)$$

$$\text{Managers} \rightarrow \text{Locations}, \text{Roles}$$

$$(\text{Managers}, \text{Locations}) \rightarrow \text{Contact Details}$$

$$(\text{Managers}, \text{Roles}) \rightarrow \text{Duties}, \text{Goals}.$$

(b) Selection of communication media: After gaining insights about group awareness, selecting the correct mode of communication is crucial for effective communication among stakeholders in a GSD environment.[43] We use a tree-based approach to select the most appropriate communication mode, mechanism, and tool for different communication tasks (refer to Figure 2).

In Figure 2, the communication tree is divided into 5 levels, where level-0 refers to the communication aspects, level-1 to the different communication tasks, level-2 to the available communication modes, level-3 to the list of available communication mechanisms, and level-4 to the list of available communication tools. The selection process begins with the identification of communication tasks, such as discussion and transfer/exchange information among stakeholders. Depending on the nature of the communication tasks, suitable modes of communication will be identified for them.

In level-2 to level-4, the communication mode, mechanism, and tool will be identified in 2 steps. In step 1, only those nodes will be considered, where the number of responses satisfying a certain threshold value is greater than the number of responses below the threshold value; otherwise, the next node at the same level of the selection tree will be processed. Mathematically, the node consideration procedure is represented by Equation 1:

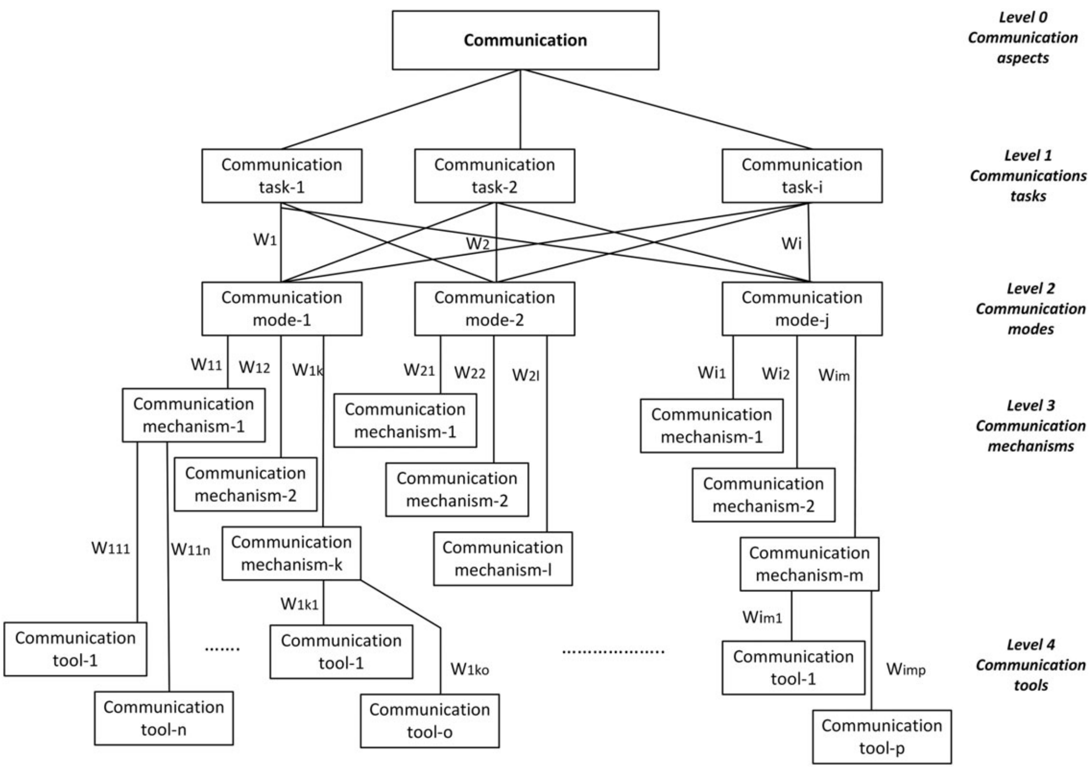

FIGURE 2 Selection of communication media in a Global Software Development environment

$$\text{Node consideration} = \sum_{\substack{i=1 \\ RC \geq \alpha}}^{n} RC_i > \sum_{\substack{i=1 \\ RC < \alpha}}^{n} RC_i, \tag{1}$$

where RC is the response count and α is the threshold value. The response count is calculated by adding the number of responses that satisfies α; however, the value of α is defined by the level of comfort on communication modes, mechanisms, and tools. Data about levels of comfort will be obtained from the Stakeholder Data Collection form (refer to Appendix S1).

In step 2, the nodes satisfying Equation 1 will be further analyzed by applying the linear compensatory model for them.[59] Mathematically, the model is represented by Equation 2:

$$\text{Individual\_rating} = \sum_{i=0}^{n} Ai, \tag{2}$$

where A_1, A_2, A_3, ... A_n belong to A (attributes).

At each level, the node with a higher mean value will be selected. All other nodes in the selection tree will be processed similarly. Thus, by taking into consideration the level of comfort demonstrated by stakeholders on different communication modes, mechanisms, and tools, the path of the tree from the root to a leaf node will then be used to identify those communication modes, mechanisms, and tools that will best meet the needs of the stakeholders.

2.2.5 | Time zones

Reducing the time required for software development is one of the main advantages of GSD. By working across different time zones, a considerable amount of development time and revenue can be saved. However, ensuring clear communication takes place between different groups of stakeholders whose working hours have little or no overlap is a challenge. Some of the problems of time zones in a GSD environment are due to (a) physical distance between stakeholders[26,27] and (b) miscommunication between stakeholders.[47] To minimize the problems that could arise because of different time zones, we suggest the use of intersection operation.

Let $T = T_1, T_2, \ldots T_n$ represent the business hours for different groups of stakeholders on weekdays. To clarify if an overlap exists between business hours, intersection operation will be performed on the elements of set T by applying Equation 3.

$$T = \{T_1 \sqcap T_2 \sqcap \ldots T_n\}. \tag{3}$$

From the results of Equation 3, overlap in working hours will be evident. If little or no overlap is found, then the following questions will be asked of those groups of stakeholders for which no overlap exists.

What will be the most suitable time for communication among the following:

(1) Monday to Friday from 9:00 AM to 5:00 PM;
(2) Monday to Friday after 5:00 PM, if required;
(3) Saturday and Sunday from 9:00 AM onwards, if required;
(4) Public holidays from 9:00 AM onwards, if required.

Once the data in relation to the abovementioned questions have been gathered, the intersection operation will be performed again to identify the mutually agreed time slot for communication between stakeholders.

2.2.6 | Impact of GSD issues on the nonfunctional requirements of a project

The functional requirements of a project are determined by the business objectives of a company and hence are independent of the location where the project is to be developed, be it locally or overseas. We are thus of the opinion that the functional requirements remain the same if the project is to be developed within a GSD environment. However, the nonfunctional requirements of a GSD project will be affected by some of the GSD issues. For instance, in the event of a system failure, could a client expect an offshore vendor to fix the problem as quickly as an onshore vendor could? Stakeholders will be informed of the impact of GSD issues on the nonfunctional requirements of a GSD project. Nonfunctional requirements include the following: localizability—possibility and capability to make changes because of regional and geographical changes; modifiability—possibility to add undetermined functionalities in future; evolvability—possibility to support new/latest technologies; composability—possibility to develop a software system by plug-and-play components; reusability—possibility to reuse components of the software system in future; security—safety aspects of a software system; integrity—trustworthiness of a remotely built software system; correctness and reliability—accuracy and dependability of a software system; maintainability—possibility of maintaining a software system from offshore sites; privacy—confidentiality feature of software system; usability—capability and training required to use a software system; performance—amount of time required to generate throughput, transition delays, and latency time, which could be interrupted sometimes by offshore sites for testing and maintenance purposes; and cost—possibility of increase in operational, maintenance, and development costs, because of changes in requirements.[60]

2.3 | Post-education assessment

Before initiating the processes of requirements elicitation and analysis, stakeholders will be assessed in relation to the knowledge gained in stage 2.2. The primary objective of this stage is to ensure that stakeholders are sufficiently aware of the GSD issues to be able to minimize the number of conflicts that could arise later during a project development. The process begins by asking stakeholders the following questions: (Q1) which of the issues could occur because of the implications of time zones, cultural diversity, knowledge management, and communication barriers on different aspects of a GSD project? The issues include the following: misunderstanding between different groups of stakeholders; lack of trust; delays in reaching agreement; difficulties in knowledge transfer; differences in organizational cultures; use of different terminologies to represent a similar set of requirements; time pressure; conflicting and diverging goals of stakeholders; knowledge clarification; language barriers; changes in requirements; and dissimilar communication processes; and (Q2) which nonfunctional requirements will be affected because of the implications of time zones, cultural diversity, knowledge management, and geographical distance between development sites in a GSD project? The nonfunctional requirements include the following: localizability; modifiability; evolvability; composability; reusability; security; integrity; correctness and reliability; maintainability; privacy; usability; performance; and cost.

Stakeholders will be awarded 1 point for each correct answer about GSD. The information thus obtained will be processed by calculating the arithmetic mean, initially for each respondent and later for each group of stakeholders. The values of arithmetic means will vary between 0 and 1, where 0 indicates "no awareness," and 1 "completely aware." The results will be examined with respect to the following threshold: (i) individual/group score < 0.5 = no knowledge of GSD; (ii) 0.5 ≤ individual/group score < 0.6 = very less knowledge of GSD; (iii) 0.6 ≤ individual/group score < 0.7 = average knowledge of GSD; (iv) 0.7 ≤ individual/group score < 0.8 = good knowledge of GSD; and (v) 0.8 ≤ individual/group score < 1.0 = excellent knowledge of GSD. The threshold for individual and group scores may vary in different GSD setups, but the threshold in this thesis is set to 0.7 and greater to assure a reasonable knowledge of GSD.

The overall results will be used as a guide for determining whether further clarification on certain issues of GSD is required. If the results show that no such clarification is necessary, the process of requirements elicitation will begin.

2.4 | Requirements elicitation and analysis

Requirements will be collected and analyzed in 4 steps: step 1—gathering requirements via story cards; step 2—identification of WH questions from stories; step 3—identification of scenarios from WH questions; and step 4—identification of use cases from scenarios (refer to Figure 3). The process continues until all the requirements have been defined.

2.4.1 | Gathering requirements using story cards

The requirements elicitation process begins by asking the client to write stories about their WANTS in relation to the prospective system. The concept of story writing has been taken from the study of Patel and Ramachandran [61] and modified. The story writing approach is used because of the following reasons: it is easy to gather information about WH questions (ie, what, who, why, and where) in a simple and concise way; mostly, the stories are written by clients, which facilitate software teams to spend more time with the client (users) and better understand the WANTS (ie, what does the client want, why do they want, when do they want, and where it will be used); and finally, the stories are easy to communicate between the different groups of stakeholders.[1]

At this instance, information regarding some of the nonfunctional requirements will also be gathered with respect to the time zone and distance between different sites of GSD. To avoid repetition, story cards will be analyzed, and related responses will be grouped by sorting similar stories into classes, discarding repetitions, and giving an appropriate name to each class. Later on, the grouped stories will be analyzed to identify possible inconsistencies, which will be resolved by contacting the client.

[1]http://www.serpicodev.com/blog/2013/03/benefits-user-stories/

Once the inconsistencies are removed, the stories will be examined to ensure completeness. We cover completeness in the point as to whether each story provides information about the following: the purpose of the story; the description of the story; the story points; concerned stakeholders; acceptance test; series of dependent and alternative actions; limitations; nonfunctional requirements; and final remarks.[61] In the case of missing information, the client will be contacted, and the required information will be collected from them.

2.4.2 | Identification of WH questions from stories

The information obtained from the story cards will be used to identify WH questions (ie, what does the client wants, why does the client wants, when it is required, and who will use it) about the functional and nonfunctional requirements of the prospective systems (refer to Table 2). The pool of WH questions will be analyzed, and a set of WH diagrams will be generated.

Later on, each WH diagram will be examined to ensure completeness. We refer to completeness in relation to whether each WH diagram provides the appropriate information: what does the client want; why does the client want; when it is required; who will use it; the chain of dependent and alternating questions; the set of limiting questions; nonfunctional requirements; and acceptance criterion. In the case of missing information, the client will be contacted, and the required information will be collected from them.

Once the gathered requirements are transformed into WH diagrams, scenarios will be identified from them. Each scenario will provide information about a particular aspect of the client's project. From the pool of identified scenarios, details about use cases will be extracted from them. In the following, we present the transformation rules, based on which requirement artifacts will be converted from 1 form to another (ie, WH diagram to scenarios and scenarios to use cases).

2.4.3 | Identification of scenarios from WH diagrams

Scenarios will be generated from WH diagrams by following the transformation rules (refer to Table 3). Information about different aspects (such as name and purpose of scenario, completion deadline, stakeholders' information, information flow, dependency between scenarios, constraints, optional and alternative flows, nonfunctional requirements, and expected results) will be extracted from WH diagrams to identify a complete scenario. Likewise, all the possible scenarios will be identified.

Similar to the stories and WH analysis, each scenario will be examined to ensure completeness. We refer to completeness in relation to whether each scenario provides the appropriate information: the scenario's name; stakeholders' information; constraints; nonfunctional requirements; information, alternative, and optional flows; approval criterion; expected results; and dependency between scenarios.[62] In the case of missing information, the client will be contacted, and the required information will be collected from them.

2.4.4 | Identification of use cases from scenarios

Use cases will be identified from the scenarios defined previously by following the transformation rules (refer to Table 4). For this reason,

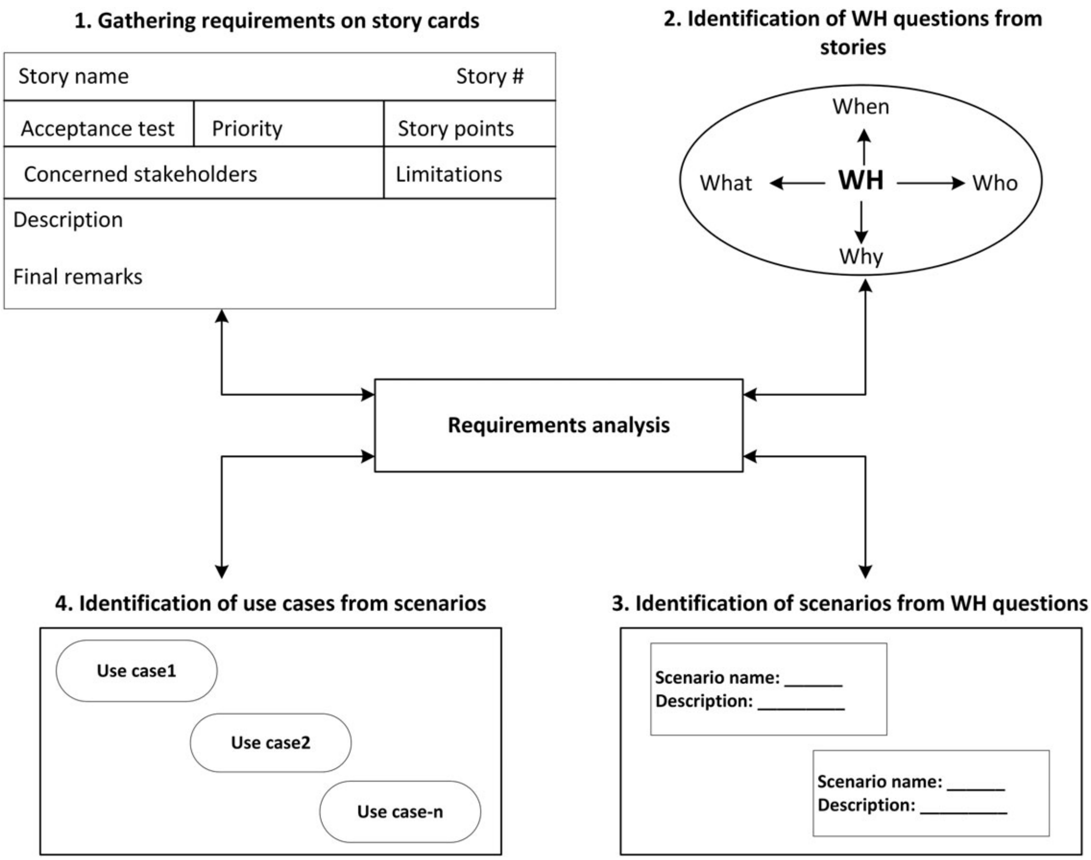

FIGURE 3 Process of requirements elicitation and analysis

TABLE 2 Transformation rules for converting stories into WH questions

Stories	WH Questions
Purpose of story	What does the client wants
Description of story	Why does the client wants
Story points	When it is required
Concerned stakeholders	Who will use it
Acceptance test	Acceptance criterion
Series of dependent actions	Chain of dependent questions
Limitations	Limiting questions
Nonfunctional requirements	Nonfunctional requirements
Series of alternative actions	Alternating questions
Final remarks	What and why do you want it

TABLE 3 Transformation rules for converting WH questions into scenarios

WH Questions	Scenarios
What does the client wants	Name
Why does the client wants	Purpose
When it is required	Completion deadline
Who will use it	Stakeholders' information
Chain of dependent questions	Information flow of actions
Limiting questions	Constraint
Nonfunctional requirements	Nonfunctional requirements
Alternating questions	Optional and alternative features
Acceptance criterion	Approval criterion
What and why do you want it	Expected results

information about the set of actors, their requirements for goals (ie, use cases), and the relationship between them is identified from the available pool of information.[13]

Following the use case identification process, each use case will be examined to ensure completeness. We refer to completeness in relation to whether each use case provides the appropriate information: goal of the use case; actors involved in the use case; associated functional requirements; list of pre/post conditions; main and alternative flows; relationship between use cases; and the acceptance criteria for each use case.[62] In the case of missing information, the client will be contacted, and the required information will be collected from them.

Finally, the completeness of nonfunctional requirements will be examined. In the study of Pressman,[63] a checklist of properties is presented that should be satisfied to ensure the completeness of nonfunctional requirements. These properties include the following: whether each nonfunctional requirement has an associated functional requirement; whether each nonfunctional requirement is achievable in a technical environment; whether each nonfunctional requirement is testable after implementation; and whether all nonfunctional requirements are noncontradictory with other requirements. Therefore, the list of nonfunctional requirements gathered in earlier steps will be analyzed for these properties with respect to different geographical locations of the teams.

TABLE 4 Transformation rules for converting scenarios into use cases

Scenarios	Use Cases
Scenario's name	Name of a use case
Stakeholders' information	Actor
Constraints	Precondition
Information flow	Main flows
Alternative and optional flows	Alternatives
Approval criterion	Approval condition
Expected results	Post conditions
Dependency between scenarios	Relationship between use cases

3 | THE OSS CASE STUDY

Client XYZ is located in Australia and has been involved in the merchandising business for the last several years, selling products to the Australian market. The client considers the needs of the customer to be very important and wants to ensure a positive relationship exists with them. The client's motive is "to sell reliable and quality products to a broad range of customers at affordable prices." Because of team work, dedication, a clear focus, and well-defined marketing strategies, the client holds a respectable position in the Australian merchandising arena.

Client XYZ wants to develop an OSS for their organization. In the shopping system, the following features are required:

- Facilitate customers/end users in the following: purchasing different products; tracking their orders; viewing sellers' information; and making payments via a secure payment platform.
- Facilitate XYZ in the following: selling different products; managing information about customers (ie, shoppers) and wholesale merchandisers (ie, sellers); managing all the orders made by the customers; and managing information about the products sold or still available.
- Easy-to-use graphical user interface.

3.1 | The software organization

ALPHA is a software organization that designs and delivers a broad range of information technology solutions including software development, hardware and software installations, network maintenance and management, desktop support and maintenance, and computer system upgrades. The main site (headquarters) of ALPHA is located in Australia, and the offshore sites are located in India and China. Since its beginning, ALPHA has proven itself to be a highly committed organization, which delivers quality information technology solutions at affordable prices.

To be able to gain the benefits of GSD, the managers of XYZ contacted a software development organization, ALPHA, to discuss their requirements for developing the OSS. Five managers from the client's organization and 8 from ALPHA (2 project analysts, 3 requirements engineers, and 3 designers) took part in the requirements elicitation and analysis. The clients, the project analyst, and the requirements engineers of ALPHA are located in Australia, and the designers reside in India.

4 | APPLYING OUR METHOD TO THE CASE STUDY

Past researchers used student groups in a university environment to play the roles of stakeholders in experiments in GSD studies.[37–40] Hence, we simulated the development of GSD requirements of XYZ using our method by creating a virtual environment for GSD within the vicinity of La Trobe University, Australia. In this virtual environment, the roles of the stakeholders were played by a group of students. With the limitations of a controlled GSD experiment, the geographical difference was simulated by ensuring that the students who performed the roles of the client, requirements engineer, and analyst have never met the students who performed the role of the designer and can only talk via the identified communications tools, thus simulating the difference in time zone.

We selected 13 undergraduate and graduate students from the Department of Computer Science and Computer Engineering, La Trobe University, Australia, and divided them into 4 teams. The teams consisted of 5, 2, 3, and 3 students who played the roles of the clients, the analysts, the requirements engineers, and the designers, respectively. Basic knowledge about the experimentation process, their roles and responsibilities, and the list of things that was expected from them was given to them via information sessions. Interaction between different student teams was performed via a set of synchronous and asynchronous collaboration tools. In the following subsections, we describe how our method was implemented using the student groups.

4.1 | Data collection

To initiate the process of devising requirements, members of the analyst team distributed data collection forms to the members of the client, requirements engineer, and design teams. A deadline of 5 working days was given to each stakeholder to complete and return the form to the members of the analyst team. After receiving the responses from all the stakeholders, the project analysts analyzed the collected data and identified any potential conflicts (refer to Table 5). The results are summarized as follows: (1) language—all the stakeholders use the English language to communicate with each other, but the way they speak, write, listen, and read English usually differs between members; (2) culture—the participants in the client and RE teams and the design teams belong to Anglosphere and Sinosphere cultural backgrounds, respectively, which could cause a cultural conflict between the members of these teams; (3) time zones and working hours—there is a 4.5-hour difference in time zones and working hours between the "client and requirements engineer teams" and the "design teams"; (4) job description—5 participants from the client organization and 3 members from the requirements engineer and design teams took part during requirements elicitation and analysis; (5) collaborative tools—all stakeholders could access the collaborative tools in their organization, but the level of comfort differed among themselves[2]; (6) learning capability—to be able to inform stakeholders about the aspects in which they required training, learning strategies were required; and

[2]Because of space limitations, we do not mention details about the first and second languages and the level of comfort in different communication modes, mechanisms, and tools in Table 4.

TABLE 5 Data collected from the stakeholders

GSD Aspect	Questions	Response	Final Remarks
Language	Common language found Number of stakeholders who speak, write, read, and listen to the common language	English C(5), R(3), D(3)	Comfort level in relation to the English language differs among stakeholders
Culture and location	Cultural backgrounds of stakeholders Geographical location of stakeholders	C/R(Anglosphere), D(Sinosphere) C and R (Australia), D(India)	Cultural differences exist between teams
Time zones and working hours	Time zones in which stakeholders work Number of overlap working hours between (R and C) and (R, C, and D)	C and R(GMT + 10), D(GMT + 5.30) Between R and C = 8 h Between R, C, and D = 3.5 h	4.5 h time difference between C, R, and D sites
Job description	Role performed during RE in GSD	C(5), R(3), D(3)	5, 3, and 3 participants will take part during elicitation and analysis
Collaborative tools	Number of stakeholders with access to communication tools Tools used for communication	C(5), R(3), D(3) C(video sessions, IM, emails, telephone) R and D(video/audio sessions, forums, emails, IM, telephone)	All the stakeholders can access and use collaborative tools. However, level of comfort differs between them
Learning capability	Desire to learn new techniques Learning strategies found to be common among stakeholders	C(5), R(3), D(3) Document ideas, rehearsal, elaboration	All the stakeholders are willing to learn new techniques
Awareness of GSD	Number of stakeholders engaged previously in GSD Number of stakeholders who are aware of - communication issues in GSD - loss of intellectual property in GSD - different government rules in GSD - distinct organizational policies in GSD - dissimilar software approaches in GSD	C(0), R(0), D(0) C(0), R(0), D(0) C(0), R(0), D(0) C(0), R(0), D(0) C(0), R(0), D(0) C(0), R(0), D(0)	None of the participants from the client, requirements engineer, and design teams have an awareness of GSD

Abbreviation: GSD, Global Software Development; RE, requirements engineering.

C = client, R = requirements engineers, D = designers.

(7) awareness of GSD—none of the participants from the client, requirements engineer, and design teams have background knowledge of GSD, because this is the first time they have taken part in a geographically distributed project.

4.2 | Educating stakeholders about GSD issues

To minimize the number of potential conflicts, the members of the analyst team educated the stakeholders about the different aspects of GSD and their impact on nonfunctional project requirements by conducting 6 information sessions in a videoconferencing environment. It took approximately 30 minutes to complete each information session. The number of information sessions mainly depends on stakeholders' knowledge about the different aspects of GSD and could be combined depending on the availability of involved stakeholders.

In the first session, the project analysts informed the stakeholders about the issues they may face because of cultural diversity, variation in time zones, communication barriers, and difficulties in knowledge management (refer to Table 6). By explaining the contents of Table 6, the analysts explained which challenge would be present during each phase of GSD.

TABLE 6 Issues faced during GSD

Issues	Communication	Culture	Knowledge Management	Time Zone
Misunderstandings	X	X	X	-
Lack of trust	X	X	X	X
Delays in reaching agreement	X	X	X	X
Knowledge transfer	X	X	X	X
Differences in organizational cultures	X	X	X	-
Use of different terminologies	X	X	X	-
Time pressure	X	X	X	X
Conflicting and diverging goals	X	-	X	-
Knowledge clarification	X	X	X	X
Variation in time zones	X	-	X	-
Changes in requirements	X	-	X	-
Dissimilar communication processes	X	X	X	X

Abbreviation: GSD, Global Software Development.

In the second session, the analysts informed the stakeholders how distance, time zone, culture, and knowledge management can affect the nonfunctional requirements of the GSD project (refer to Table 7). For instance, development work will be conducted in India, but localizability is required for the Australian stakeholders; offshore support will be required after the project is completed; if plug-and-play components are used, then only those components that do not require licensing should be used, otherwise, the clients need to be informed; despite the fact that there is period of 4.5 working hours shared commonly both by the Australian and Indian teams, it could be possible that interaction is required after business hours; it could be possible that end users will not be able to submit their requests to the offshore teams because of system nonavailability; and the Indian team will be required to travel to Australia to give training to the stakeholders/end users in Australia.

After providing insights about the issues and their impact on software development, the project analysts informed the stakeholders about the different cultural contexts involved in running the case study. Therefore, the analysts explained the differences between the Sinosphere and Anglosphere cultures in the third session by using Lewis' cross-cultural model (refer to Table 8).[48]

In the fourth session, the analysts explained the ways by which stakeholders can minimize the knowledge management issues. For this purpose, they defined a set of keywords that will be used during the development process of the online system (refer to Table 9).

In the fifth session, the analysts explained the ways by which stakeholders can overcome communication barriers. To clarify organizational roles in a GSD environment, the analysts gathered organizational details of the client, requirements engineers, and design teams. For demonstration purposes, we have presented the details of the client team in Table 10. Likewise, details of the requirements engineer and design teams were also recorded.

To further facilitate communication between stakeholders, the analysts identified the communication mode, mechanisms, and tools for different types of communication tasks by using Equations 1 and 2, which could be easily used by the stakeholders. The communication tasks identified in this case study are primarily related to project discussion and knowledge transfer/exchange among stakeholders. During the session, the analysts obtained stakeholders' feedback from the Stakeholder Data Collection form (refer to Appendix S1) and analyzed the data in 3 steps. In step 1, they identified communication modes for each communication task. In Table 11, we provided details about the identification of communication modes for the client team to engage in project discussions. In Tables 11–13, c stands for "client" and a "attributes." The attributes are a_1 = ease-of-use; a_2 = reliability; a_3 = performance; a_4 = cost-benefit analysis; and a_5 = security. Each attribute was graded over a scale of 1 to 4, where 1 represents "not at all comfortable," 2 "less comfortable," 3 "comfortable," and 4 "completely comfortable." The threshold value is set to 3, so that all the stakeholders must feel comfortable in communication modes, mechanisms, and tools.

For the identified communication mode, the analysts identified the communication mechanisms and tools that would be used for communication purposes by the client team in steps 2 and 3 (refer to Tables 12 and 13).

Similar to steps 1 to 3, the analysts identified the communication modes, mechanisms, and tools for each group of stakeholders (refer to Table 14).

Finally, in the sixth session, the analysts identified the business hours in which different groups of stakeholders can interact and discuss project issues by computing the intersection operation for the business hours of the client, requirements engineer, analyst, and design teams.

Let "T" represent the working time,

$$T_1(\text{Client}) = 9{:}00 \text{ am to } 5{:}00 \text{ pm, GMT} + 10, \text{Monday-Friday}$$
$$T_2(\text{Requirement Engineers}) = 9{:}00 \text{ am to } 5{:}00 \text{ pm, GMT} + 10, \text{Monday-Friday}$$
$$T_3(\text{Analyst}) = 9{:}00 \text{ am to } 5{:}00 \text{ pm, GMT} + 10, \text{Monday-Friday}$$
$$T_4(\text{Design}) = 9{:}00 \text{ am to } 5{:}00 \text{ pm, GMT} + 5.30, \text{Monday-Friday}.$$

TABLE 7 Impact of GSD aspects on the nonfunctional requirements of a GSD project

	GSD Aspects			
Nonfunctional Requirements	**Distance**	**Culture**	**Knowledge Management**	**Time Zones**
Localizability	X	X	X	X
Modifiability	-	-	X	-
Evolvability	-	-	X	-
Composability	-	X	X	-
Reusability	-	X	X	-
Integrity	X	X	X	X
Correctness and reliability	-	-	X	X
Privacy	X	X	X	-
Maintainability	X	X	X	X
Usability	X	X	X	X
Performance	X	X	X	X
Security	X	X	X	X
Cost	X	X	X	X

Abbreviation: GSD, Global Software Development.

TABLE 8 Cultural differences between the Australian and Indian teams

	Cultural Influence	Cultural Differences: Australian Teams (Anglosphere Culture)	Cultural Differences: Indian Team (Sinosphere Culture)	RE Activities Affected
Communication	Style of communication	Polite and direct	Polite and indirect	Gathering Analysis Management
	Medium of communication	Written	Verbal	
	Communication agenda	Attach to agenda	Mostly go for repeats	
	Use of body language	Less body language	Slight body language	
	Purpose of speech	Information	To encourage agreements	
	Understanding written and verbal communication	Quick response	Slow response	
	Communication pauses	Prefer short pause	Like long pause	
	Time spent on communication	Talks half of the time	Listen mostly	
Working habits	Sources of data collection	Statistics and research	People and data source	Gathering Analysis Specification Validation Management
	Planning of things	Planning ahead	Consider general facts	
	Involvement of logic	Deal with logic	Do not deal with logic	
	Division of work	Classify projects	Consider whole project	
	Interruptions during work	Hardly interrupt	Do not interrupt	
	Style of doing work	Task-oriented	People-oriented	
	Nature of commitments	Attach to facts	Attach to statements	
	Use of diplomacy	Truth then diplomacy	Diplomacy then truth	
	Lack of patience	A little bit impatient	Mostly patient	
	Respect toward organization	Respect organization	Respect networks	
	Style of living	Split experts with social	Join experts with social	
	Multitasking of activities	One thing at a time	Respond to actions	
	Time commitment	Timekeeping is vital	Timekeeping is important	
	Work compromise	To complete work	For future relationships	
	Value of contract	Firm	Negotiable	
	Organizational goal	Focus on short-term profit	Rely more on long-term profit	

Abbreviation: RE, requirements engineering.

TABLE 9 List of keywords used in the online system

Ontology Type	Keywords
Domain	E-business, online system
Generic	Buyers account, friend referral, marketing, navigation, payment gateway, security, sellers account, shopping cart, traffic
Representational	-
Task	Bulk emails, consistency, cost, customer recommendation, easy buying, edit products, finalizing order, inbound links, multiple products, navigation location, newspapers and television advertisements, password, payment authentication, payment criteria, payment processing, privilege, promotions via email, recommendation based on previous buying history, referral number, referred product, running banners, sales tax, secured channel, security certificate, special offers, specific to customer behavior, user name, website registration at different platforms
Method	Account generation, friend referral, marketing, navigation, payment gateway, pricing, security, shopping cart, traffic management

TABLE 10 Organizational details of stakeholders

Organization	Departments	Managers	Contact Details	Duties
Client	Information technology	Mr ABC	Australia, GMT + 10, abc@xyz.com	Technology head
	Sales/presales	Mr DEF	Australia, GMT + 10, def@xyz.com	Senior sales officer
	Marketing	Mr GHI	Australia, GMT + 10, ghi@xyz.com	Marketing manager
	Human resource	Mr JKL	Australia, GMT + 10, jkl@xyz.com	Human resource manager
	Finance	Mr MNO	Australia, GMT + 10, mno@xyz.com	Senior financial officer

By applying Equation 3, the analysts computed

$$T = \{T_1 \sqcap T_2 \sqcap T_3\} = \text{9:00 am to 5:00 pm, GMT} + 10, \text{Monday-Friday}$$
$$T = \{T_1 \sqcap T_2 \sqcap T_3 \sqcap T_4\} = \text{12:30 to 5:00 pm, GMT} + 10, \text{Monday-Friday}.$$

The analysts found that because of the client, requirements engineer, and analyst teams being in the same geographical location, there was a complete overlap between their business hours. However, there was only a 4.5-hour overlap in the business hours of the client, requirements engineer, analyst, and design teams. Therefore, different groups of stakeholders can interact within the overlapping periods of time.

4.3 | Post-education assessment

Once the education (training) process was over, members of the analyst team performed the post-education evaluation. They assessed

TABLE 11 Identification of the communication mode for project discussions

	Verbal					Written					Final Outcome (Summation of Graded Attributes by Equation 2)		
Stakeholders	a_1	a_2	a_3	a_4	a_5	a_1	a_2	a_3	a_4	a_5	Verbal	Written	Decision
c_1	4	3	3	4	3	3	3	2	2	3	17	13	Verbal
c_2	4	3	3	3	3	3	2	3	2	3	16	13	Verbal
c_3	4	3	4	3	2	3	3	3	1	2	16	12	Verbal
c_4	4	3	4	3	3	2	3	2	3	3	17	13	Verbal
c_5	4	3	3	4	3	2	3	3	1	3	17	12	Verbal

TABLE 12 Identification of the communication mechanism for project discussions

	Audio/Videoconferencing					Telephone Calls					Final Outcome (Summation of Graded Attributes by Equation 2)		
Stakeholders	a_1	a_2	a_3	a_4	a_5	a_1	a_2	a_3	a_4	a_5	Audio/Video	Telephonic Calls	Decision
c_1	4	3	4	3	3	4	3	3	2	3	17	15	Audio/video
c_2	3	3	3	4	3	4	3	4	2	1	16	14	Audio/video
c_3	4	3	3	3	2	4	3	4	1	2	15	14	Audio/video
c_4	3	3	4	4	3	3	4	3	2	3	17	15	Audio/video
c_5	3	3	4	4	2	4	3	3	2	2	16	14	Audio/video

TABLE 13 Identification of the communication tool for project discussions

	Skype					Adobe Acrobat Connect					Final Outcome (Summation of Graded Attributes by Equation 2)		
Stakeholders	a_1	a_2	a_3	a_4	a_5	a_1	a_2	a_3	a_4	a_5	Skype	Adobe Acrobat	Decision
c_1	3	3	4	3	4	3	2	2	4	3	17	14	Skype
c_2	3	3	4	4	3	3	4	2	3	2	17	14	Skype
c_3	4	3	3	4	3	2	3	2	3	4	17	14	Skype
c_4	3	3	3	4	3	2	3	3	4	1	16	13	Skype
c_5	4	3	4	3	2	3	2	3	3	2	16	13	Skype

TABLE 14 Identification of communication method for GSD stakeholders

		Group of Stakeholders			
Communication Task	Communication Aspect	Client	Requirements Engineer	Design	Choice Made
Project discussion	Mode Mechanism Tool	Verbal Audio/video Skype	Verbal Audio/video Skype	Verbal Audio/video Skype	Verbal Audio/video Skype
Knowledge transfer and exchange	Mode Mechanism Tool	Written Messaging Emails	Written Messaging Emails and instant messaging	Written Messaging Emails and instant messaging	Written Messaging Emails

Abbreviation: GSD, Global Software Development.

stakeholders in relation to the knowledge gained about the implications of time zones, cultural diversity, and language and communication barriers on different aspects of a GSD project. Each respondent is awarded 1 point for a correct answer about GSD. The information obtained is processed by calculating the arithmetic mean, initially for each respondent and later for each group of stakeholders. The values of arithmetic means vary between 0 and 1, where 0 indicates "no awareness" and 1 "completely aware." The results obtained after performing the evaluations are presented in Tables 15 and 16. Based on the results shown in Tables 15 and 16 and after evaluating the results with the defined threshold limit (ie, 0.7 and greater in this case study), the analysts recognized that the group total scores for the client team is ≈0.679 and 0.677, the requirements engineer team ≈0.83 and 0.79, and the design team ≈0.82 and 0.75, for the awareness of issues introduced by GSD and the awareness of nonfunctional requirements that are affected by GSD, respectively. As a result, most stakeholders

TABLE 15 Awareness of issues introduced by GSD

Users		Aspects					
Group	Number	Communication	Culture	Time Zone	Knowledge Management	Individual Total Score	Group Total Score
Client	1	10	9	8	9	0.75	0.679166667
	2	8	9	8	8	0.6875	
	3	9	8	7	8	0.666666667	
	4	9	8	8	7	0.666666667	
	5	8	7	7	8	0.625	
Requirements engineer	6	11	10	10	9	0.833333333	0.833333333
	7	10	10	9	10	0.8125	
	8	11	11	9	10	0.854166667	
Design	9	10	9	9	10	0.791666667	0.826388889
	10	11	10	9	10	0.833333333	
	11	10	10	10	11	0.854166667	

Abbreviation: GSD, Global Software Development.

TABLE 16 Awareness of nonfunctional requirements affected by GSD

Users		Aspects					
Group	Number	Distance	Culture	Time Zone	Knowledge Management	Individual Total Score	Group Total Score
Client	1	8	8	7	9	0.727272727	0.677272727
	2	7	7	8	8	0.681818182	
	3	8	8	7	8	0.704545455	
	4	8	8	7	7	0.681818182	
	5	6	7	6	7	0.590909091	
Requirements engineer	6	9	8	9	9	0.795454545	0.795454545
	7	8	9	8	9	0.772727273	
	8	10	10	8	8	0.818181818	
Design	9	8	8	9	9	0.772727273	0.75
	10	7	8	8	8	0.704545455	
	11	8	9	8	9	0.772727273	

Abbreviation: GSD, Global Software Development.

are aware of the effects of geographical distance, time zones, cultural diversity, and knowledge management. Therefore, the requirements elicitation process should be started.

4.4 | Requirements elicitation and analysis

To initiate the process of requirements elicitation, members of the analyst team requested the RE team members to pay a visit to the client's organization to gather an initial set of requirements from them. Consequently, the RE team members requested the 5 client managers in a face-to-face meeting to write stories about their WANTS in relation to the requirements of the OSS. To illustrate the process of requirements gathering via stories, Figures 4 and 5 show the stories collected from 2 client managers about the features of the order placement aspect of the online system.

After collecting basic information on the online system, the RE team members, together with the members of the analyst team, analyzed the stories to eliminate repetitions and identify potential inconsistencies. With reference to the requirements depicted in Figures 4 and 5, inconsistencies were found in the functional and nonfunctional requirements. For instance, variations exist in the stories regarding the order verification and time to retrieve order details. Furthermore, there are some pieces of information about nonfunctional requirements that are only present in 1 of the stories. To resolve these issues, members of the RE and analyst teams contacted the client's managers and clarified the matters with them. Based on the obtained information, the analysts and the requirements engineers created the story for the order placement part of the online system (refer to Figure 6).

Once the stories for the project requirements had been created, members of the RE and analyst teams analyzed each story to ensure completeness. Figure 7 shows detail regarding completeness evaluation of the story for the order placement part of the system.

Following the process of story analysis, the analyst and RE team members created the WH diagrams for different aspects of the OSS. Figure 8 illustrates the WH diagram for the order placement part of the online system.

Afterwards, the members of the RE and analyst teams analyzed each WH diagram to ensure completeness. Figure 9 shows detail regarding the completeness evaluation of the WH diagram for the order placement part of the system.

Once the WH diagrams for the entire project requirements had been created and analyzed, members of the RE and analyst teams started the scenario identification process. To define a scenario, they examined each WH diagram in a detailed manner and extracted the required information from them. As a result, a set of scenarios defining the different aspects of the online system was identified. Table 17 presents the details of the scenario for the order placement part of the system.

After this, the members of the RE and analyst teams analyzed each scenario to ensure completeness. Figure 10 shows the detail regarding

Story name: Order placement		**Story #:** 1
Acceptance test: Customer successfully place their order	**Priority:** High	**Story points:** With project completion
Concerned stakeholders: Customers		**Limitations:** Desired item/book is available

Description: To facilitate customers in online shopping, the "online shopping system" must be capable of accepting online orders. The system should help customers in searching desired item/book, preparation of shopping cart, verification of credentials (i.e. contact and payment details), and order finalization.

It should be compulsory for customers to verify their order details and in case of incorrect contact and/or payment details, it is required for them to update their information before placing the order. Later on, the system must update the modified information in the centralized database, process the order and supply receipt to customer.

Some other requirements in this regard are listed below.
- Customers can place order 24/7.
- The response time for system to retrieve order details should not be more than 10 seconds.
- Orders should be processed in a secure channel.
- End-users with different background knowledge can easily place order
- Regardless of time difference between Australia and other offshore sites, 24/7 support will be required
- System will be developed at offshore locations, but localizability must be done with respect to Australian traits

Final remarks: Helping customers in order placement.

FIGURE 4 Story written by the first client manager

Story name: Order placement		**Story #:** 2
Acceptance test: Order should be successfully placed and processed	**Priority:** High	**Story points:** With project completion
Concerned stakeholders: Customers		**Limitations:** Required item is available

Description: The online system should be capable of helping customers in finding desired item, processing payment details over secured channel and order finalization. The system should make it compulsory for customers to prepare shopping cart and supply and verify shipping/payment details, before they proceeds towards order finalization. In response, the system must generate an order confirmation receipt for future correspondences.

Some additional requirements are mentioned below.
- 24/7 order placement.
- System should be able to retrieve order details within 30 seconds.
- Offshore support is required.

Final remarks: Order should be successfully placed by customers and later processed by system.

FIGURE 5 Story written by the second client manager

the completeness evaluation of the scenario for the order placement part of the system.

In the next step, the members of the RE and analyst teams generated the use cases for the set of scenarios identified in the previous step. Consequently, they came up with a number of use cases that explained the working of the functional requirements in a detailed manner. In addition to use case generation, information about the nonfunctional requirements that could affect the functionality captured in the use cases was also recorded. Table 18 presents the details of the use cases for the order placement part of the system, and the set of nonfunctional requirements is listed in Table 19.

Later, members of the RE and analyst teams analyzed each use case to ensure completeness. Figure 11 shows details regarding the completeness evaluation of the use case for the order placement part of the system.

Following the process of scenario identification and use case generation, members of the RE and analyst teams contacted the members of the design team to discuss the project requirements. They organized a videoconferencing session and explained the set of use cases and the list of nonfunctional requirements to the members of the design team. During this process, all the members examined the requirements in a detailed manner. As a result, they were able to identify areas that required further clarification and attention. For instance, it was mentioned by the client that they required 6 months' support but the information provided in this regard was not clear. To address this, the RE team members

Story name: Order placement		**Story #:** 3
Acceptance test: Order should be successfully placed and processed	**Priority:** High	**Story points:** With project completion
Concerned stakeholders: Customers		**Limitations:** Desired item/book is available

Description: To facilitate customers in online shopping, the "online shopping system" must be capable of accepting online orders. The system should help customers in searching desired item/book, preparation of shopping cart, verification of credentials (i.e. contact and payment details), and order finalization.

The system should make it compulsory for customers to prepare shopping cart and supply and verify shipping/payment details, before they proceeds towards order finalization. In case of incorrect contact and/or payment details, it is required for customers to update information before placing their orders. The modified information thus obtained must be updated in the centralized database. Finally, the system must generate an order confirmation receipt for future correspondences.

Some other requirements in this regard are listed below.
- Customers can place order 24/7.
- The response time for system to retrieve order details should not be more than 10 seconds.
- Orders should be processed in a secure channel.
- End-users with different background knowledge can easily place order
- Regardless of time difference between Australia and offshore sites, 24/7 support will be required
- System will be developed at offshore locations, but localizability must be done with respect to Australian traits

Final remarks: Order should be successfully placed by customers and later processed by system.

FIGURE 6 Story for order placement

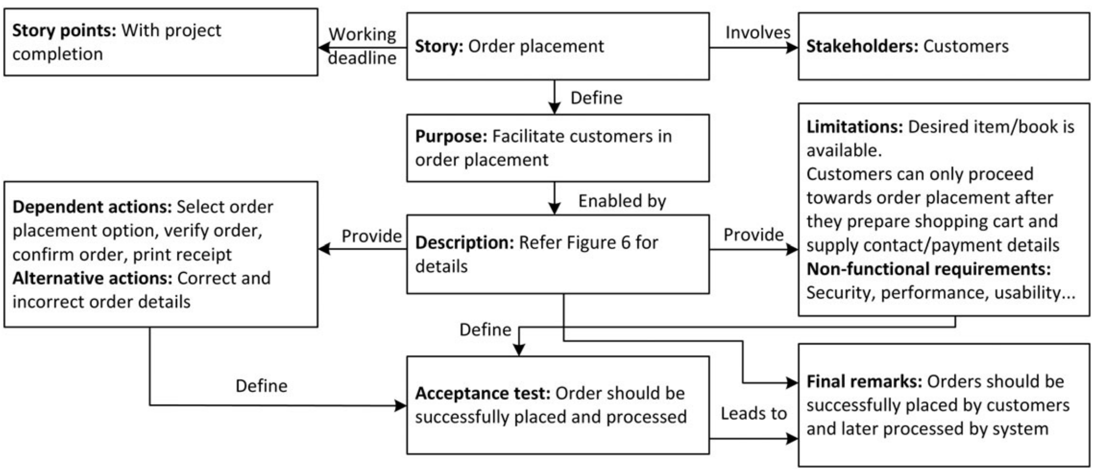

FIGURE 7 Completeness evaluation of story for order placement

organized a joint videoconferencing session for all the teams (ie, client, RE, analyst, and design) to discuss the issues identified. Based on the information gathered during the videoconferencing session, the members of the RE, analyst and design teams made appropriate changes to the existing set of requirement and finalized the details regarding the terms and conditions for the 6 months' support.

Finally, the members of the RE, analyst, and design teams analyzed the nonfunctional requirements for completeness by evaluating them with respect to the properties mentioned in Section 2.4. For the 6 nonfunctional requirements listed in Table 18, these team members considered 1 requirement at a time and evaluated it on the basis of following properties: (1) associated function requirements; (2) achievable in technical environment; (3) testable after implementation; and (4) noncontradictory with other requirements. Figure 12 shows the details regarding the completeness evaluation of 1 (out of 6) nonfunctional requirement that is associated with the order placement part of the online system. The remaining 5 nonfunctional requirements for the order placement part were also analyzed in a similar way. Later, the nonfunctional requirements for other features of the OSS were analyzed, accordingly.

5 | APPLYING THE CONVENTIONAL METHOD TO THE CASE STUDY

To evaluate the effectiveness of the proposed requirements elicitation and analysis method, the conventional RE method that does not consider GSD specifics was used for the same GSD project. To avoid

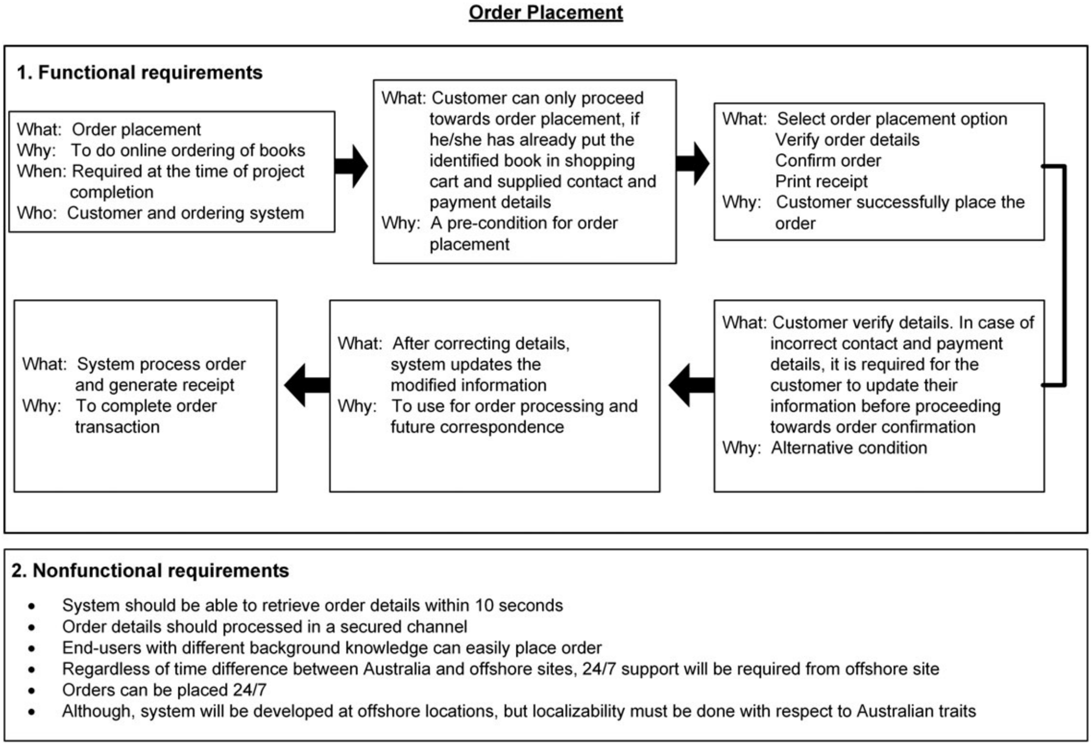

FIGURE 8 WH diagram for order placement

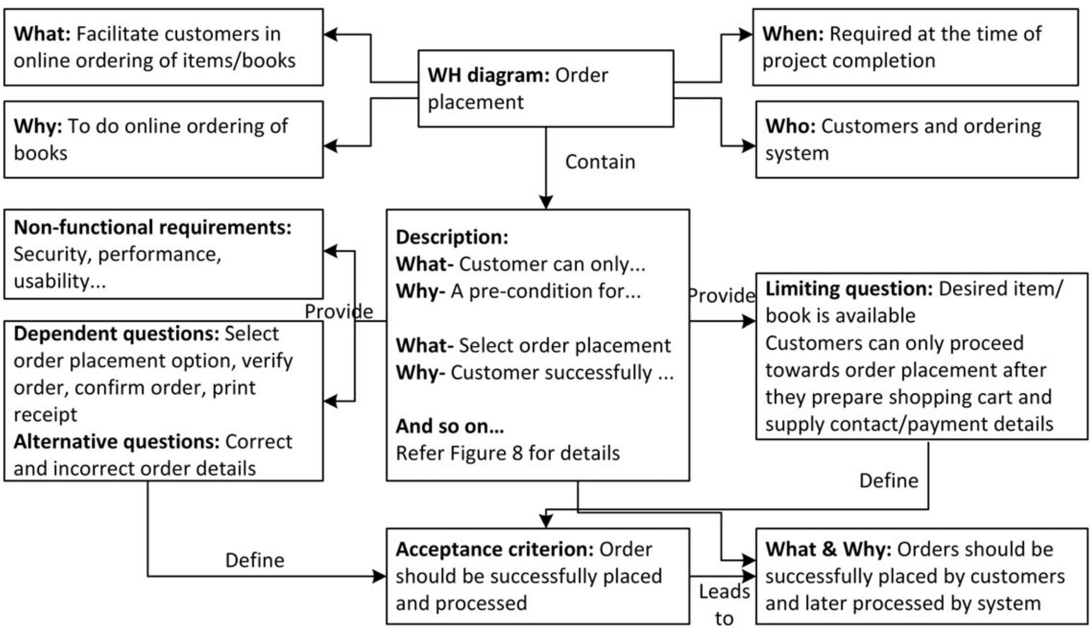

FIGURE 9 Completeness evaluation of WH diagram for order placement

learning effect, a different group of 13 undergraduate and graduate students from the Department of Computer Science and Computer Engineering, La Trobe University, Australia, was selected and later divided into 5 teams. The teams consisted of 5, 2, 3, 3, and 3 students who played the roles of the clients, the analysts, the requirements engineers, and the designers, respectively. Basic knowledge about the experimentation process, their roles and responsibilities, and the list of things that was expected from them was given to them via information sessions. Interaction between different student teams was performed by a set of synchronous and asynchronous collaboration tools.

The list of activities involved in the proposed and the conventional RE methods can be found in Table 20. Although story writing, scenarios, and use case techniques are used in the conventional method, factors like different time zones and distances are not considered when formulating requirements.

TABLE 17 Scenario description of order placement

Scenario	Description
Name	Order placement
Purpose	Online ordering of a book
Completion deadline	Required at the time of project completion
Stakeholders' information	Customer and ordering system
Dependency between scenarios	There are different types of transactions in an online shopping system, and book ordering is one of them
Information flow (ie, series of actions)	1. Select order placement option 2. Verify order details 3. Confirm order 4. Print receipt
Constraint	Customer can only proceed toward order placement if they have already put the identified book in the shopping cart and supplied their contact and payment details
Alternatives	If order details are correct, the customer confirms their order, and the system generates the transaction receipt. Otherwise, it is necessary for the customer to update their information before proceeding toward order placement
Expected results	Customer successfully places the order
Nonfunctional requirements	● System should be able to retrieve order details within 10 s ● Order details should be processed in a secured channel ● End users with different background knowledge can easily place orders ● Regardless of the time difference between Australia and offshore sites, 24/7 support will be required from the offshore sites ● Orders can be placed 24/7 ● Although the system will be developed at offshore locations, localizability must be ensured with respect to Australian traits

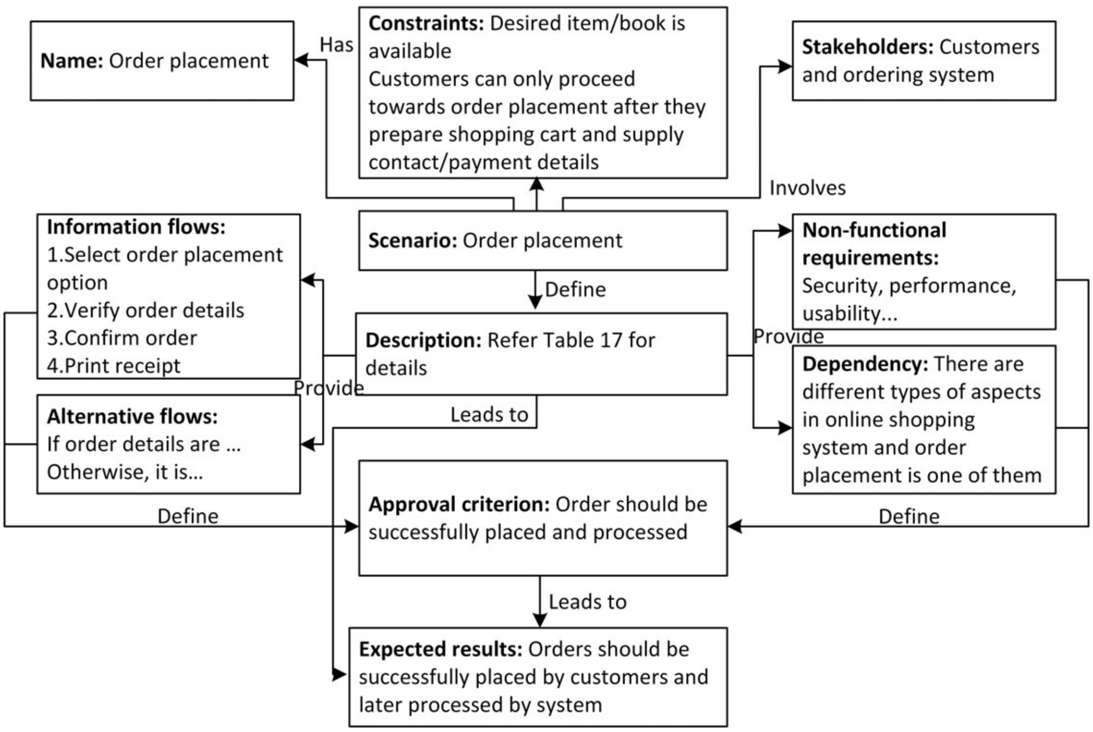

FIGURE 10 Completeness evaluation of the scenario for order placement

In the following, we describe how the student groups used the conventional RE method to perform requirements elicitation and analysis in GSD environment.

5.1 | Requirements elicitation and analysis

Depending on the nature of involvement, 5 managers from the client's organization and 8 from ALPHA (2 project analysts, 3 requirements engineers, and 3 designers) took part in the requirements elicitation and analysis process. To initiate the process of requirements elicitation, the requirements engineers visited the client and gathered information on the functional and nonfunctional requirements of the OSS using the story writing approach.[61] They requested each of the client's managers to write their WANTS about the requirements of the OSS in short stories. After completing the story writing and gathering sessions, the requirements engineers, together with the project analysts,

TABLE 18 Use case description of order placement

Use Case	Description
Name	Order placement
Goal	Online ordering of a book
Actor	Customer and ordering system
Precondition	Customer has already searched for a book and supplied the required details for ordering
Main flows	1. Customer selects the ordering option 2. The system generates order details for the customer 3. Customer verifies the order details 4. The system processes the order and generates the receipt for the customer 5. Customer takes the receipt and successfully terminates the ordering session
Alternative flows	3a. In the case of missing or wrong information, it is necessary that the customer updates their information 3b. System updates the information in its central database
Post conditions	Customer successfully performs the transaction

TABLE 19 Description of nonfunctional requirements for order placement

Nonfunctional Requirements	Description
Performance	The system should be able to retrieve order details within 10 s
Security	Order details should be processed in a secured channel
Usability	End users with different background knowledge can easily place orders
Support	Regardless of the time difference between Australia and offshore sites, 24/7 support will be required for 6 months from the offshore sites
Availability	Orders can be placed 24/7
Localizability	Although the system will be developed at offshore locations, localizability must be ensured with respect to Australian traits

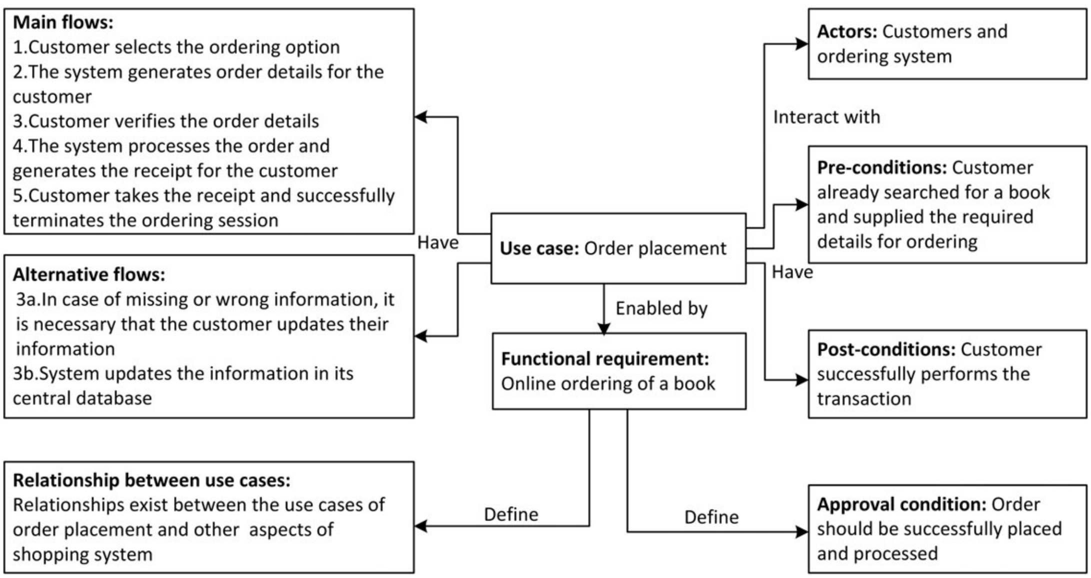

FIGURE 11 Completeness evaluation of use case for order placement

processed the gathered pieces of requirements from stories to eliminate inconsistencies and later identified a possible set of scenarios and use cases from them. In Tables 21–23, details regarding the user story, scenario, and use case that are generated for the order placement aspect of the shopping system are presented. Thereafter, Table 24 presents details on the functional and nonfunctional requirements and the set of potential scenarios and use cases for the OSS.

After completing the requirements elicitation process, the requirements engineers, the project analysts, and the designers analyzed the gathered pieces of requirements in a videoconferencing session using the traditional contract-style requirements list. To do so, they prepared a checklist of requirements to examine whether each functional requirement has an associated nonfunctional requirement, the use case was identified, the priority was defined, and comments regarding incompleteness, inconsistencies, and ambiguities were made, if required (refer to Table 25). From the results of Table 25, they came to know that there were several requirements that needed additional clarification by the client. As a result, the requirements engineers, the project analysts,

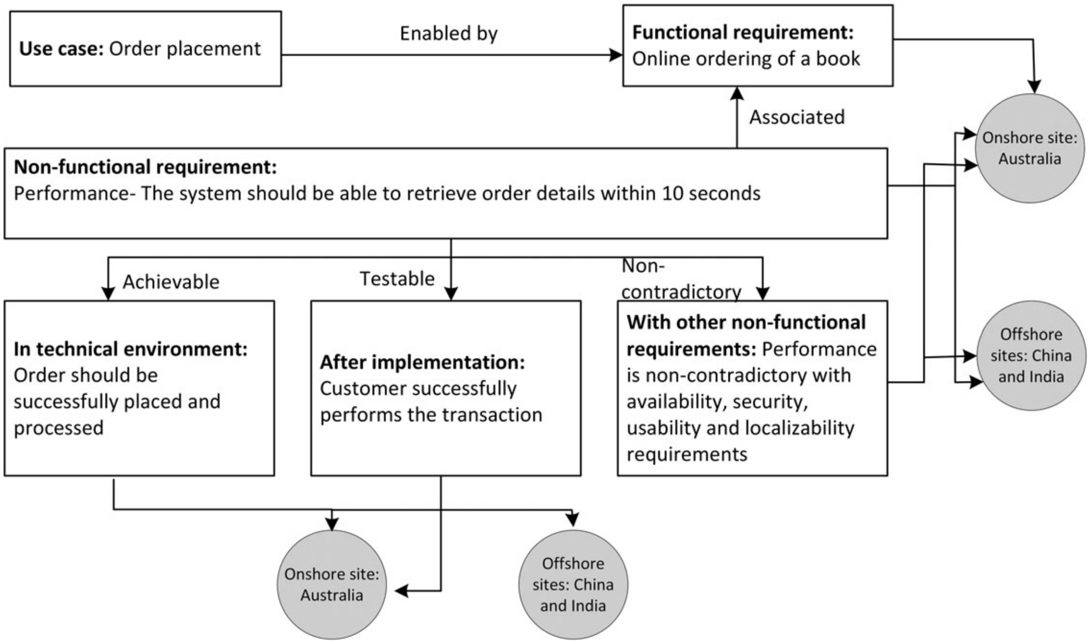

FIGURE 12 Completeness evaluation of nonfunctional requirements

TABLE 20 List of activities involved in the conventional method and the proposed RE method

Tasks	Activities	Conventional Method	Proposed Method
Data collection	Gathering details about GSD stakeholders Conflicts identification	- -	X X
Educating stakeholders about GSD	Culture, communication, knowledge management, and time zone	-	X
Post-education assessment	Evaluating stakeholders knowledge Decision regarding further clarification of knowledge	- -	X X
Requirements elicitation and analysis	Story writing (user stories) WH questions Scenarios Use cases	X X X	X X X X

Abbreviation: GSD, Global Software Development; RE, requirements engineering.

TABLE 21 Processed story of order placement

Story card	
Story card no.	15
Project name	Online shopping system
Story name	Order placement
Date	July 10, 2013
Story	Order placement shall be one of the main aspects of our online shopping system. To do so, the prospective system should be able to take orders from customers, process their credentials, and finalize their details
Acceptance test	It is compulsory that customer details should be correct to process the order successfully. Otherwise, delays in order finalization or rejection of order could be possible
Risk	High
Points to consider	The overall system should be secure, available 24/7 and fast

the designers, and the client organized a couple of meetings (videoconferencing sessions) to discuss the issues presented in Table 25.

6 | DISCUSSIONS OF THE RESULTS

After completing the requirements elicitation and analysis process for both methods, we interviewed the student teams in 5 steps about their experiences in relation to the different aspects of the used methods. In step 1, we interviewed teams about the level of usefulness of the different aspects of requirements elicitation and analysis. In step 2, we investigated the level of comfort felt by each team in working with the other teams. In step 3, we asked the teams to state the frequency of conflicts that occurred during requirements elicitation and analysis. In step 4, we interviewed the teams about the number of requirements identified and analyzed. In step 5, we interviewed the teams about the time spent on

TABLE 22 Scenario description of order placement

Scenario	Description
Name	Order placement
Purpose	To place an online order
Completion deadline	With project completion
Stakeholders' information	Customer and online system
Details	Order placement shall be one of the main aspects of our online shopping system. To do so, the prospective system should be able to take orders from customers, process their credentials, and finalize their details It is compulsory that customer details should be correct to process the order successfully. In case of incorrect or missing information, customer should be prompted to provide correct details Otherwise, delays in order finalization or rejection of order could be possible
Desired result	Order shall be placed successfully
Nonfunctional requirements	Performance: The time to retrieve order details should be fastSecurity: Order details should be processed and stored securelyAvailability: Order should be placed 24/7

TABLE 23 Use case description of order placement

Use case	Description
Name	Order placement
Goal	To place an online order
Actor	Customer and online system
Precondition	Customer searched the desired book
Main flows	1. The order selection menu is selected 2. The system take order from the customer 3. The system verifies the order details 4. The system generates the order receipt
Alternative flows	3.1. If the provided information is missing or wrong, the system should notify the customer to re-supply the required information
Post conditions	Order should be placed successfully

different activities of requirements elicitation and analysis. Finally, we analyzed and discussed the data obtained during these steps.

6.1 | Usefulness of the RE methods

To determine the level of usefulness of the proposed and the conventional requirements elicitation and analysis methods, we contacted members of the client, requirements engineer, analyst, and design teams to rate the different activities (also called aspects) of the used method on a scale of 1 to 4, where 1 indicates "not useful," 2 "less useful," 3 "useful," and 4 "very useful," based on their level of involvement in the various aspects. To analyze the data gathered in this step, we used arithmetic mean to determine the level of usefulness (refer to Table 26).

From the results shown in Table 26, it can be seen that overall, the respondents from different teams rated the proposed requirements elicitation and analysis as either useful or very useful for the accomplishment of the abovementioned aspects. The primary reasons why our method was considered useful than the conventional RE method are as follows: (1) it enabled the early identification and minimization of conflicts related to cultural diversity, working in different time zones, language barriers, and difficulties in communication and coordination, and it helped members of different teams to work and collaborate with different groups of stakeholders at multiple geographical locations; (2) the consideration of nonfunctional requirements with respect to different time zones and increased distance helped the teams to think about the underlying requirements in a broader domain, which is often neglected in collocated software development;

TABLE 24 List of functional and nonfunctional requirements gathered using the conventional RE method

Requirement No.	Functional Requirement	Nonfunctional Requirement	Scenario id	Use case id
1	Service component	Performance, safety, security, availability, reliability	S-001ONS	UC-001ONS
2	Purchase goods	Performance, safety, security, availability	S-002 ONS	UC-002 ONS
3	Browse catalogue	Performance, availability	S-003 ONS	UC-003 ONS
4	Select product	Performance, availability	S-004 ONS	UC-004 ONS
5	Make payment	Performance, availability, safety, security, reliability	S-005 ONS	UC-005 ONS
6	Order placement	Performance, security, availability	S-006 ONS	UC-006 ONS
7	Supply order details	Performance, availability, reliability, security	S-007 ONS	UC-007 ONS
8	Order tracking	Performance, security, safety	S-008 ONS	UC-008 ONS
9	Information about seller	Performance, availability	S-009 ONS	UC-009 ONS
10	View seller information	Availability	S-0010 ONS	UC-0010 ONS
11	Payment component	Performance, safety, security, availability, reliability	S-0011 ONS	UC-0011 ONS
12	Payment procedure	Performance, security, safety, availability, reliability	S-0012 ONS	UC-0012 ONS
13	Payment using credit card	Performance, security, safety, availability, reliability	S-0013 ONS	UC-0013 ONS
14	Supply card details	Security, safety, availability, reliability	S-0014 ONS	UC-0014 ONS
15	Authentication mechanism	Performance, security, availability, reliability	S-0015 ONS	UC-0015 ONS

Abbreviation: RE, requirements engineering.

TABLE 25 Requirements analysis via contract-style requirements list

Requirement No.	Functional Requirement	Nonfunctional Requirement	Use Case ID	Priority	Additional Comments
1	Service component	Performance, safety, security, availability, reliability	UC-001ONS	High	Not applicable
2	Purchase goods	Performance, safety, security, availability	UC-002 ONS	High	Additional information is required
3	Browse catalogue	Performance, availability	UC-003 ONS	Medium	Not applicable
4	Select product	Performance, availability	UC-004 ONS	Medium	Not applicable
5	Make payment	Performance, availability, safety, security, reliability	UC-005 ONS	High	Provided information is not sufficient
6	Order placement	Performance, security, availability	UC-006 ONS	High	Not applicable
7	Supply order details	Performance, availability, reliability, security	UC-007 ONS	High	Not applicable
8	Order tracking	Performance, security, safety	UC-008 ONS	High	Ambiguities in order tracking
9	Information about seller	Performance, availability	UC-009 ONS	Low	Information is missing about sellers
10	View seller information	Availability	UC-0010 ONS	Low	Not applicable
11	Payment component	Performance, safety, security, availability, reliability	UC-0011 ONS	High	Not applicable
12	Payment procedure	Performance, security, safety, availability, reliability	UC-0012 ONS	High	Not applicable
13	Payment using credit card	Performance, security, safety, availability, reliability	UC-0013 ONS	High	Not applicable
14	Supply card details	Security, safety, availability, reliability	UC-0014 ONS	High	Additional information is required
15	Authentication mechanism	Performance, security, availability, reliability	UC-0015 ONS	High	Unclear information about authorization mechanism

TABLE 26 Level of usefulness of different aspects of the requirements elicitation and analysis method

Different Aspects	*N*	GSD-RE	Conventional RE
Thinking about requirements	13	3.38	2.5
Describing requirements	13	3.53	2.5
Analyzing requirements	8	3.62	2.25
Negotiating requirements	13	3.38	1.91
Finalizing requirements	13	3.38	1.75

Abbreviation: GSD, Global Software Development; RE, requirements engineering.

and (3) the requirements elicitation and analysis method, which used story writing, WH questions, scenarios, use cases, and analysis procedure, with respect to different time zones and distances, was easy to use and helped the stakeholders to think, describe, analyze, negotiate, and finalize requirements better in globally distributed perspective.

6.2 | Level of comfort working with other teams

To determine the level of comfort student teams felt in working with other teams, the members of the client, requirements engineer, analyst, and design teams were asked to rate their level of comfort on a scale of 1 to 4, where 1 indicates "not comfortable," 2 "less comfortable," 3 "comfortable," and 4 "very comfortable," based on their level of involvement in the GSD project. To analyze the data gathered in this step, we used the arithmetic mean to determine the level of comfort.

6.2.1 | Client vs other teams

Table 27 presents details regarding the level of comfort the client team expressed on the other teams in the proposed and the conventional methods of RE.

6.2.2 | Requirements engineers vs other teams

Table 28 presents details regarding the level of comfort the requirements engineers expressed on the other teams in the proposed and the conventional methods of RE.

6.2.3 | Analysts vs other teams

Table 29 presents details regarding the level of comfort the analysts expressed on the other teams in the proposed and the conventional methods of RE.

TABLE 27 Level of comfort expressed by the client team on the other teams

	N	GSD-RE	Conventional RE
Requirements engineers	5	3.2	1.6
Analysts	5	3	1.4
Designers	5	2.6	1

Abbreviation: GSD, Global Software Development; RE, requirements engineering.

TABLE 28 Level of comfort expressed by the requirements engineers on the other teams

	N	GSD-RE	Conventional RE
Client	3	3.66	1.66
Analyst	3	4	1.66
Designers	3	3.33	1.33

Abbreviation: GSD, Global Software Development; RE, requirements engineering.

TABLE 29 Level of comfort expressed by the analysts on the other teams

	N	GSD-RE	Conventional RE
Client	2	3.5	2
Requirements engineers	2	4	2
Designers	2	3.5	1.5

Abbreviation: GSD, Global Software Development; RE, requirements engineering.

6.2.4 | Designers vs other teams

Table 30 presents details regarding the level of comfort the designers expressed on the other teams in the proposed and the conventional methods of RE.

The main reason for the levels of comforts, expressed in Tables 27–30, is mainly due to the knowledge provided to all the teams about the different aspects of GSD, such as cultural diversity, language barriers, knowledge management, and differences in time zones. With this knowledge, the GSD teams had appropriate expectations of the other teams and were then able to communicate and discuss the project better with each other. In addition, the identification and finalization of the appropriate communication modes, mechanisms, and tools for different types of communication tasks before initiating the process of RE, together with the technical contribution made in the GSD-RE method, helped the GSD teams in establishing and maintaining a higher level of comfort with the other teams.

6.3 | Frequency of conflicts

In step 3, we asked each team to state the frequency of conflicts that occurred during the different activities of RE in relation to the proposed and conventional RE methods on a scale of 1 to 4, where 1 indicates "rarely," 2 "occasionally," 3 "less frequently," and 4 "very frequently." To analyze the data obtained during this step, we applied the arithmetic mean (refer to Table 31).

From the results shown in Table 31, it can be seen that overall, the respondents from the different teams indicated less conflicts during requirements elicitation and analysis using the proposed method, mainly because of the following: (1) prior knowledge of cultural, temporal, social, and geographical aspects of GSD ensured that teams had appropriate expectations of each other that assisted the communication and coordination process; (2) the identification of suitable communication modes, mechanisms, and tools for communication before initiating the processes of requirements elicitation and analysis helped the teams in minimizing the frequency of conflicts during requirements thinking, discussion, negotiation, analysis, and finalization; and (3) the consideration of nonfunctional requirements with respect to different time zones and increased distance helped teams to think about the underlying requirements of a software system in a broader domain.

TABLE 30 Level of comfort expressed by the designers on the other teams

	N	GSD-RE	Conventional RE
Client	3	2.33	1.33
Requirements engineers	3	3.33	1.33
Analysts	3	3.33	1.33

Abbreviation: GSD, Global Software Development; RE, requirements engineering.

TABLE 31 Frequency of conflicts that occurred during different aspects

Different aspects	*N*	GSD-RE	Conventional RE
Thinking about requirements	13	1.7	3.38
Describing requirements	13	1.61	3.38
Analyzing requirements	8	1.37	3.62
Negotiating requirements	13	1.61	3.69
Finalizing requirements	13	1.61	3.46

Abbreviation: GSD, Global Software Development; RE, requirements engineering.

6.4 | Number of requirements identified and analyzed

In step 4, we interviewed the relevant student teams in relation to the number of requirements identified and analyzed using the proposed and the conventional method of requirements elicitation and analysis. In Table 32, details regarding the number of functional and nonfunctional requirements are presented. From the results, it can be seen that the students who implemented the proposed method of requirements elicitation and analysis identified/analyzed 22 functional and 8 nonfunctional requirements, compared to 17 functional and 5 nonfunctional requirements using the conventional method of requirements elicitation and analysis.

6.5 | Time spent on different activities of RE

In step 5, we interviewed the relevant student teams in relation to the time spent on the different activities of requirements elicitation and analysis using the proposed and conventional methods of RE. In Table 33, details regarding the time spent on educating stakeholders about GSD are presented. From the results, it can be seen that the students who implemented the proposed RE method spent 2.5 hours to gain knowledge about the different aspects of GSD. However, no education on GSD was given to the students in the conventional method of RE.

In Table 34, details regarding the post-education assessment are presented. It can be seen from the results that it took 2 hours to assess the knowledge of students regarding the different aspects of GSD using the GSD-RE method. Because of the fact that no education on GSD was provided to the students in the conventional RE method, therefore, no post-education assessment was performed.

TABLE 32 Details on the number of elicited requirements

Number of Requirements Generated	GSD-RE	Conventional RE Method
Functional requirements	19 + 3 = 22	15 + 2 = 17
	19 before making changes	15 before making changes
	3 after making changes	2 after making changes
Nonfunctional requirements	8	5

Abbreviation: GSD, Global Software Development; RE, requirements engineering.

TABLE 33 Time spent on educating stakeholders about GSD

Activities	Time Spent Using the GSD-RE	Time Spent Using the Conventional RE Method
Culture	30	Did not educate stakeholders about GSD
Knowledge management	30	
Communication	30	
Time zones	30	
Impact of GSD on nonfunctional requirements	30	
Total time, min	150	0
Total time, h	2.5	0

Abbreviation: GSD, Global Software Development; RE, requirements engineering.

TABLE 34 Time spent on post-education assessment regarding GSD

Activities	Time Spent Using the GSD-RE	Time Spent Using the Conventional RE Method
Asked questions	60	Did not perform post-education assessment
Evaluate knowledge	60	
Total time, min	120	0
Total time, h	2	0

Abbreviation: GSD, Global Software Development; RE, requirements engineering.

In Table 35, details regarding the time spent on the different activities of requirements elicitation and analysis are presented. It can be seen that the students who implemented the proposed method of requirements elicitation and analysis took 330 minutes to gather and analyze user stories, 176 minutes to generate and analyze WH questions, 154 minutes to identify and analyze scenarios, 176 minutes to generate and analyze use cases, and 300 minutes to evaluate requirements for completeness and consistency that makes the total of 1136 minutes (18.93 hours). However, the other groups of students who implemented the conventional RE method took 374 minutes to gather and analyze user stories, 255 minutes each to identify scenarios and use cases, and 450 minutes to analyze requirements for completeness and consistency, giving a total of 1334 minutes (22.23 hours).

6.6 | Summary of the results

After analyzing the results presented in the earlier sections, the following observation is made about the students who used the conventional RE method.

- Because of the lack of knowledge about GSD and the non-consideration of increased distances and time zones during requirements elicitation and analysis, the student/GSD teams faced challenges during requirements thinking, describing, analysis, negotiation, and finalization. As a result, they identified a subset of overall requirements and missed out some of the requirements of the OSS.

TABLE 35 Time spent on requirements elicitation and analysis

Activities	Time Spent Using the GSD-RE	Time Spent Using the Conventional RE Method
User stories	15 × 22 = 330	19 × 17 = 374
WH questions	8 × 22 = 176	Did not generate
Scenarios	7 × 22 = 154	15 × 17 = 255
Use cases	8 × 22 = 176	15 × 17 = 255
Completeness and inconsistencies evaluation	Functional requirements = 150 Nonfunctional requirements = 150	Functional requirements = 450 Nonfunctional requirements = 0
Total time, min	1136	1334
Total time, h	18.93	22.23

The student teams expressed a lower level of usefulness of the different aspects of the conventional RE method, felt less comfortable working with other student teams in a GSD environment, stated the increased frequency of conflicts that occurred during the fundamental activities of requirements elicitation and analysis, and spent more time to complete the assigned tasks.

In comparison to the findings made about the students who used the conventional RE method, the following observations are made about the students who used the proposed method of requirements elicitation and analysis.

- Having acquired knowledge on cultural diversity, language barriers, differences in time zones, and their impact on nonfunctional requirements, the students (GSD teams) had appropriate expectations of each other and were able to better communicate and discuss the project with each other. Understanding nonfunctional requirements with respect to different time zones and increased distance helped students think about the underlying requirements in a broader domain.
- The story writing, WH questions, scenarios, and use case techniques are used to gather requirements for a collocated software development project, whereby factors like different time zones and distances need not be considered. These techniques are used in our method in ways that help GSD teams think about requirements with respect to different time zones and distance.

The students rated different aspects of the proposed method as either useful or very useful for GSD, expressed a higher level of comfort working with other teams, and the frequency of conflicts that occurred during the elicitation and analysis process was comparatively lower than the conventional RE method. Considering the fact that students performed additional activities to prepare themselves for GSD, they finish the entire process of requirements elicitation and analysis in a little bit more time, but developed a comparatively complete set of requirements, than the groups of students who implemented the conventional RE method. Although, these judgments are based on the findings presented in Sections 4 and 5, it could still be applied to other similar GSD projects.

6.7 | Threats to validity

The findings presented in Sections 4 and 5 have the following validity threats. The results were obtained from a controlled experiment, where students played the roles of the different groups of stakeholder instead of actual stakeholders in an industrial environment; and the present findings were supported by straight forward statistics because of the small sample size that we had, but instead they should be supported by a statistical analysis through having a bigger sample size.

None of the participants have previous knowledge of GSD; the results obtained in this experimental setup cannot be generalized for stakeholders having some knowledge of GSD.

We have selected undergraduate and postgraduate students from the Department of Computer Science and Information Technology, La Trobe University. The difference in educational qualification can be a selection bias threat. The experiments lasted for several months; events like examination pressure, work commitments, and personal issue occurred during this period that could affect participant's behavior and performance.

When applying our method to the case study, the students were from 1 university. Although we selected students from different cultural backgrounds, the work involving students from different universities with dissimilar cultural backgrounds needs to be examined. Situational specifics (eg, location, time, supervision, and role of investigator) can potentially limit the generalizability of the results. Researcher's expectations and experiment bias can also be a threat to validity.

7 | COMPARISON WITH THE RELATED WORK

In the literature, there are only a very few papers published on GSD requirements elicitations and analysis. Most of them are not directly related to our work. In the study of Solis and Ali,[64] the collaboration tool "Spatial Hypertext Wiki" was presented to support the different processes of RE. Wiki provides a shared virtual place, where geographically distributed stakeholders can share/view artifacts, discuss and communicate RE issues, resolve prioritization conflicts, and devise solutions for unresolved issues under discussion, however, no GSD issues were addressed. The authors in one study[65] proposed a cognitive-based method to address technology selection issues in requirements elicitation. Their method uses fuzzy logic to identify appropriate elicitation methods and techniques, in accordance with the cognitive features of involved stakeholders. No GSD requirements elicitation technique was proposed. Sabahat et al[66] conducted a survey on the existing methods and techniques of GSD requirements elicitation and the challenges associated with them. Based on the survey findings, they only suggested performing requirements elicitation and analysis iteratively, until a reasonable set of requirements is obtained but did not present any work done in this area. Chang et al[67] focused on the scenarios in which stakeholders mostly use informal notations to describe their requirements. For such types of situations, an agent-based architecture, collection of elicitation approaches, and the ways by which formal explanations of informal requirements can be elicited are presented. In several studies,[64,65,67] information regarding cultural and temporal aspects of GSD is missing. None of this work is directly related to ours.

A study conducted by Aranda et al[43] is closer to our work. They attempted to minimize the number of conflicts by educating stakeholders about a few aspects of GSD. However, the following aspects are not covered in their work.

- How to construct an ontology and to what extent the use of ontologies facilitated the authors in minimizing the language differences between the stakeholders.
- How to identify a suitable time for communication across multiple time zones.
- Impact of GSD issues on the nonfunctional requirements (eg, localizability, performance, modifiability, and evolvability) of a project. Because of a lack of awareness of these impacts, stakeholders

usually do not consider these factors when undertaking requirements elicitation and analysis, which in turn, affects later stages of project development.

- A criterion to assess the level of conflict minimization and to decide if further information sessions are required before initiating the processes of requirements elicitation and analysis.
- A method of requirements elicitation and analysis for GSD projects.

We therefore addressed these missing aspects in our work. As a result, our work is different from the study of Aranda et al[43] in the following ways.

- To reduce language barriers, we developed an ontology method to clarify knowledge among stakeholders.
- To have effective communication among stakeholders, we developed an organizational ontology to obtain information about location, contact details, roles, and duties of each stakeholder in a GSD environment. Furthermore, we suggested a tree-based approach to determine the appropriate communication mode, mechanism, and tool for different communication tasks.
- To identify a suitable time for communication across different time zones, we presented the use of the intersection operation.
- In our method, we explained to the stakeholders how the nonfunctional requirements of a GSD project will be affected by some of the GSD issues.
- We defined a criterion to evaluate the level of conflict minimization among stakeholders.
- We devised a method of requirements elicitation and analysis to facilitate stakeholders in requirements gathering and analysis.

8 | CONCLUSIONS AND FUTURE WORK

In conducting requirements elicitation and analysis, an ineffective or a lack of communication between stakeholders presents a significant problem. The problem is even more severe for a GSD project because of additional factors like cultural difference, language and communication barriers, difficulties in knowledge management, and differences in time zones. Taking into consideration the factors involved in GSD, the ways by which requirements are captured and analyzed in collocated software development cannot be used effectively in a GSD environment. To date, there are very few research papers describing the work done on GSD requirements elicitation and analysis. In this paper, we have presented GSD-RE, a method for requirements elicitation and analysis for GSD. Our method consists of 4 stages: (1) data collection; (2) educating stakeholders about GSD issues; (3) post-education assessment; and (4) requirements elicitation and analysis. Through stages 1 to 3, the "stakeholders" were given knowledge of GSD issues like cultural diversity, language barriers, knowledge management, and differences in time zones. With this knowledge, the "stakeholders" have the right expectations of each other and then were able to communicate and discuss the project better with each other.

We have applied our method to a case study of an OSS, where the roles of client, requirements engineer, project analyst, and designers were played by a group of students. It helped resolve and minimize the number of possible issues regarding cultural diversity, language barriers, different time zones, knowledge management, and communication between the stakeholders. Nonfunctional requirements that took into consideration of factors like difference in time zones and increased distance necessitated the "stakeholders" to think about the underlying requirements in a broader domain, which are often neglected in a collocated software development project. The story writing, WH questions, scenarios, and use case techniques were used to gather requirements, whereby factors like different time zones and distances were considered and the GSD teams were put in a situation to think more about requirements in a GSD environment.

Further, we used a conventional requirements elicitation and analysis method that does not consider GSD specifics for the same GSD project, so that a comparison between the results could be made. The results showed that our method helped minimize the number of problems or conflicts among the "stakeholders" when forming requirements in a GSD environment; and a set of higher quality requirements was developed. We can say that the results indicated the potential benefits of our method over the conventional method. Our method is especially useful for those groups of stakeholders who are taking part in GSD for the first time or are not aware of the fundamental aspects and issues of RE in GSD.

A weakness of the validation of our method is that our findings were not supported by a statistical analysis and hence, it should be treated as a preliminary validation. To increase the confidence in our method, we need to have a more solid validation by finding a commercial partner who would collaborate with us in experimenting it in a real-life environment.

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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